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Advanced Computer Architecture

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Name
Last Name
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You'll have 15' to complete the exam

Problem 1 (25% aka 1pt)	
Problem 2 (25% aka 1pt)	
Problem 3 (25% aka 1pt)	
Problem 4 (25% aka 1pt)	
Total (100%)	

Problem 1
Which of the following statements about branch prediction are FALSE?
We may have more than a single FALSE statement. Only correct answers will be counted for the score.

Answer 1: The behavior of each branch is always assumed independent from other branches (**FALSE**)
Answer 2: Branch prediction can only happen at compile time (**FALSE**)
Answer 3: Different techniques can be combined in order to improve prediction accuracy
Answer 4: Profile-Driven is an example of a static branch predictor

Answer 1
Different techniques make different assumptions in order to make a prediction, for example an
n-bit-BHT only considers the history of the branch to be predicted (with n regulating how much of the past history to consider).
An (m,n) Correlating Branch predictor on the other hand, takes into account the behavior of the last m branches.
Branch prediction can happen both at compile time (static) or during the execution (dynamic), the two approaches we briefly described above are examples of dynamic branch prediction techniques while Profile-Driven or Delayed Branch are static predictors.
Static predictions may be used to assist dynamic predictors.

Problem 2
The "Branch Always Taken" technique is not applicable to the five-stage MIPS pipeline! Is this statement true or false?

Answer1: **True**
Answer2: False

Answer 2

Because we used “applicable” in the question, we considered both the answers as correct. It is applicable, but it does not make sense. Therefore the “right” answer was True (but we considered correct also false, as said, because it is applicable)
It is true, in this technique as soon as the branch is decoded, and the target address is computed, we assume the branch to be taken and begin fetching and executing at the target. Because in the five-stage MIPS pipeline we don’t know the target address any earlier than we know the branch outcome, there is no advantage to this approach.

Problem 3
What is the reason a 1 bit-BHT is usually outperformed by a 2 bit-BHT?
We may have more than a single TRUE statement. Only correct answers will be counted for the score.

Answer 1: A 2-bit BHT leverages more information about the history of a branch (**TRUE**)
Answer 2: The use of 2 bits allows to index a larger number of branch instructions
Answer 3: The difference in performance is not significant
Answer 4: The use of 2 bits improves the accuracy of the associated BTB

Answer 3
A 2-bit BHT needs two mispredictions in order to change prediction. This means it also leverages more of the past history of the branch in order to make its predictions. The improvement is clear for example in the case of two nested loops where the mispredictions on the inner loop are halved with respect to the 1bit-BHT.

Problem 4
In the context of branch hazards, which of the following assertions are FALSE?
We may have more than a single FALSE statement. Only correct answers will be counted for the score.

Answer 1: The Early Evaluation of PC behaves in the same way regardless of the preceding instructions (**FALSE**)
Answer 2: The Early Evaluation of PC computes branch target address during the EX phase (**FALSE**)
Answer 3: With only forwarding and stalling, it requires two clock cycles until the branch decision is taken
Answer 4: The Early Evaluation of PC could need stalls in some particular cases

Answer 4

1. If a comparison register is a destination of preceding ALU instruction or 2nd preceding load instruction a further stall is needed, two in the case of an immediately preceding load instruction.
2. The target address is computed during the ID phase.
3. The result is available in the register after the EX phase of the branch instruction.
4. For what stated before.